Biochemistry Cumulative Exam October 6, 2000

A rat is fed a "high protein" diet consisting only of aspartate, and small amounts of vitamins and minerals.

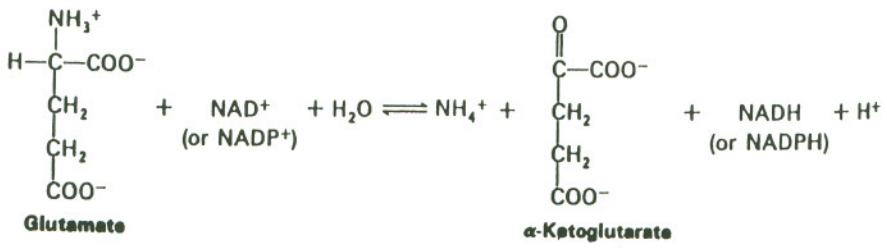
1. (15) Will the rat be in a state of ketosis? Will the rat lose weight according to the Atkins diet?

2. (30) Can glucose be synthesized in the liver of the rat using aspartate as the sole source of carbon and energy? If so, give an enzymatic pathway for the synthesis of glucose from aspartate in the liver cell, assuming only low, catalytic concentrations of other metabolites.

3. (30) Write a complete balanced equation for the process in question 2, giving the number of O_2, CO_2, urea, and ATP generated or consumed.

4. (25) If the rat is fed aspartate labeled with C-14 at carbon 3, which carbon(s) of glucose will first be labeled?

Aspartate: $^{-}OOC\text{-}CHNH_3^{+}\text{-}^{*}CH_2\text{-}COO^{-}$

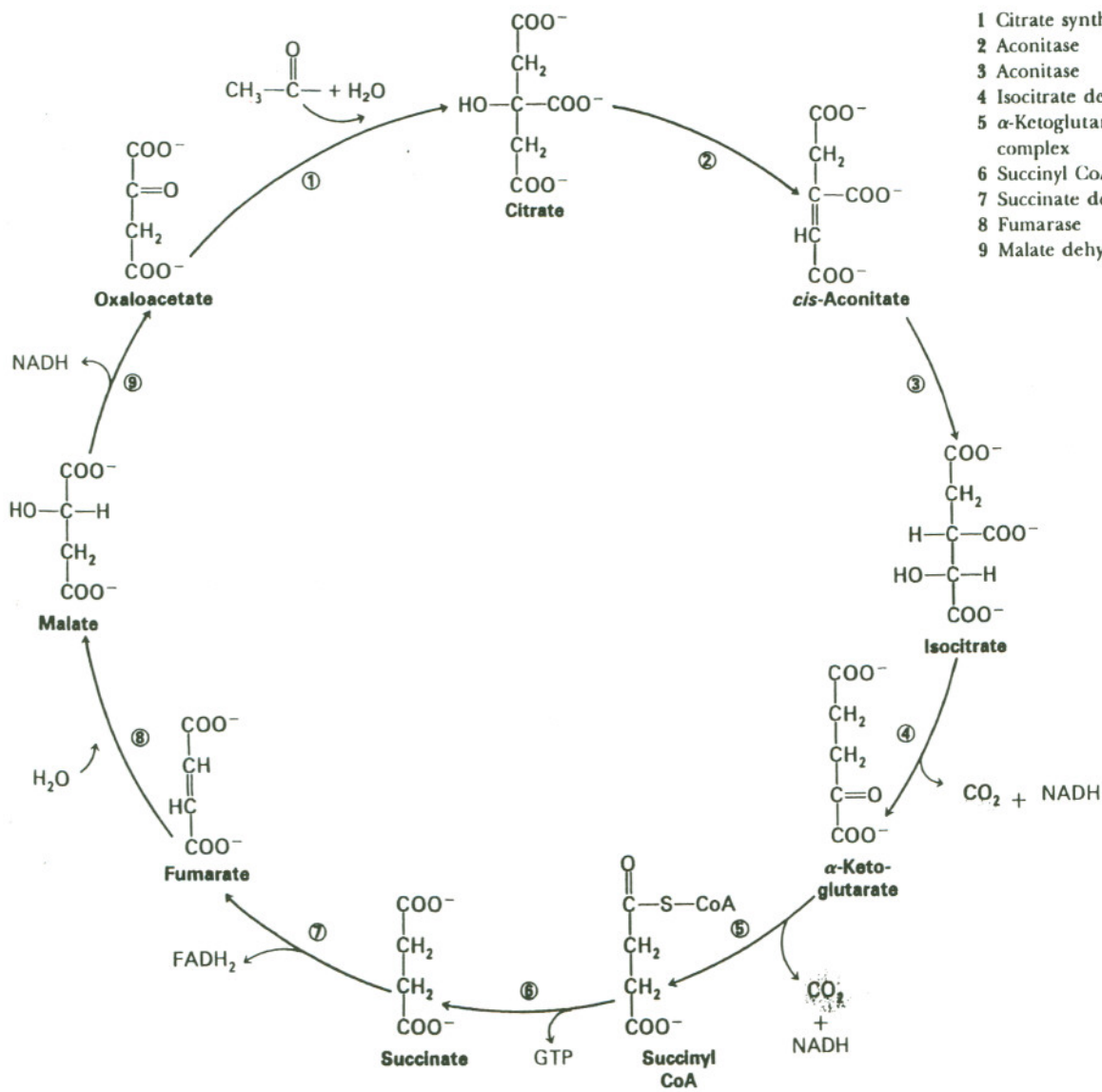


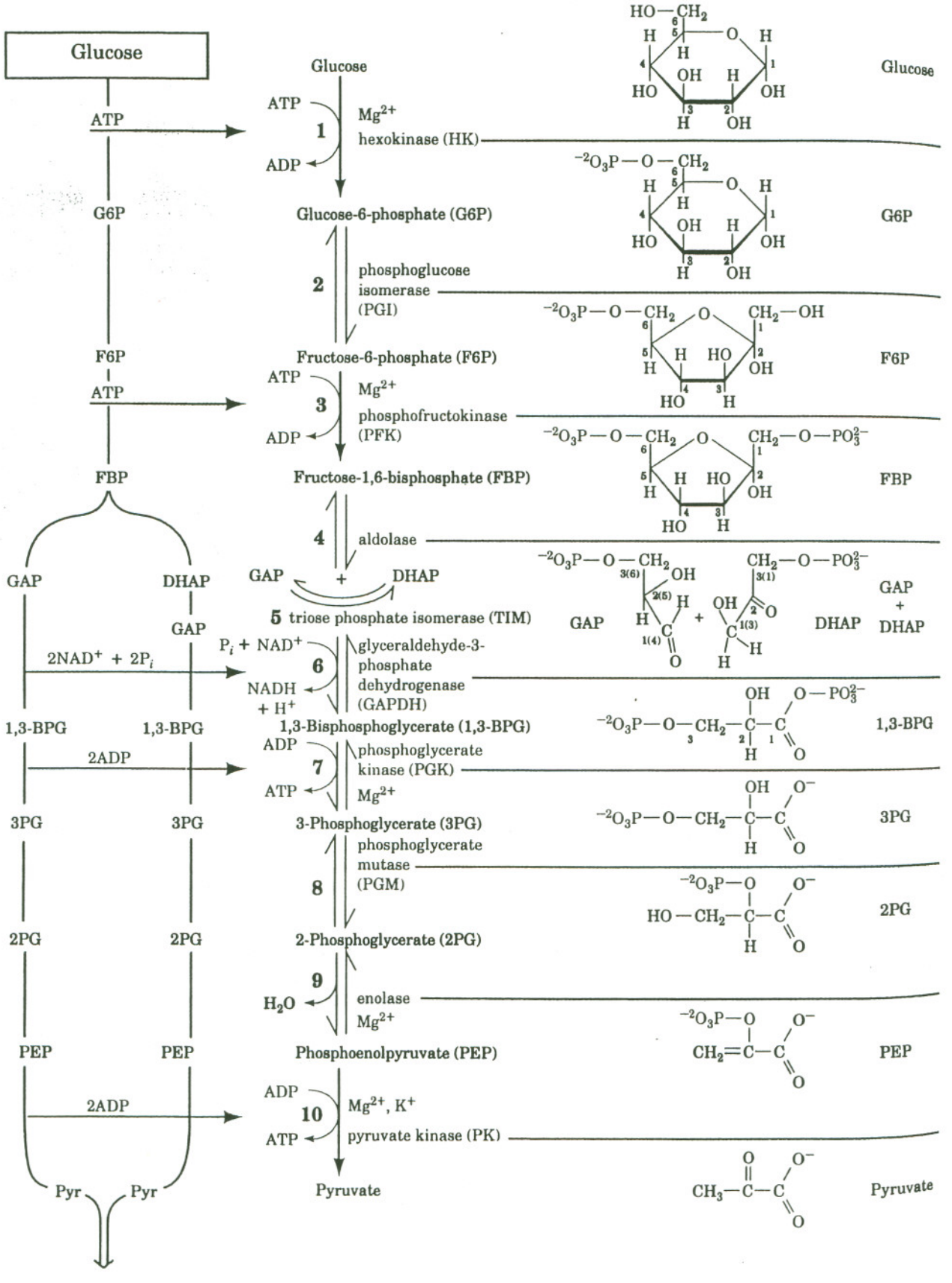
The stoichiometry of urea synthesis is

$$CO_2 + NH_4^{+} + 3\ ATP + aspartate + 2\ H_2O \longrightarrow urea + 2\ ADP + 2\ P_i + AMP + PP_i + fumarate$$

following enzymes, as numbered in the diagram:

1 Citrate synthase
2 Aconitase
3 Aconitase
4 Isocitrate dehydrogenase
5 α-Ketoglutarate dehydrogenase complex
6 Succinyl CoA synthetase
7 Succinate dehydrogenase
8 Fumarase
9 Malate dehydrogenase




Glucose
ATP
G6P
F6P
ATP
FBP
GAP
DHAP
GAP
2NAD+ + 2Pi
1,3-BPG
1,3-BPG
2ADP
3PG
3PG
2PG
2PG
PEP
PEP
2ADP
Pyr
Pyr
Glucose
ATP
1
Mg2+
hexokinase (HK)
ADP
Glucose-6-phosphate (G6P)
2
phosphoglucose isomerase (PGI)
Fructose-6-phosphate (F6P)
ATP
3
Mg2+
phosphofructokinase (PFK)
ADP
Fructose-1,6-bisphosphate (FBP)
4
aldolase
GAP + DHAP
5 triose phosphate isomerase (TIM)
Pi + NAD+
6
glyceraldehyde-3-phosphate dehydrogenase (GAPDH)
NADH + H+
1,3-Bisphosphoglycerate (1,3-BPG)
ADP
7
phosphoglycerate kinase (PGK)
ATP
Mg2+
3-Phosphoglycerate (3PG)
8
phosphoglycerate mutase (PGM)
2-Phosphoglycerate (2PG)
9
H2O
enolase
Mg2+
Phosphoenolpyruvate (PEP)
ADP
10
Mg2+, K+
pyruvate kinase (PK)
ATP
Pyruvate
Glucose
G6P
F6P
FBP
GAP
DHAP
GAP + DHAP
1,3-BPG
3PG
2PG
PEP
Pyruvate

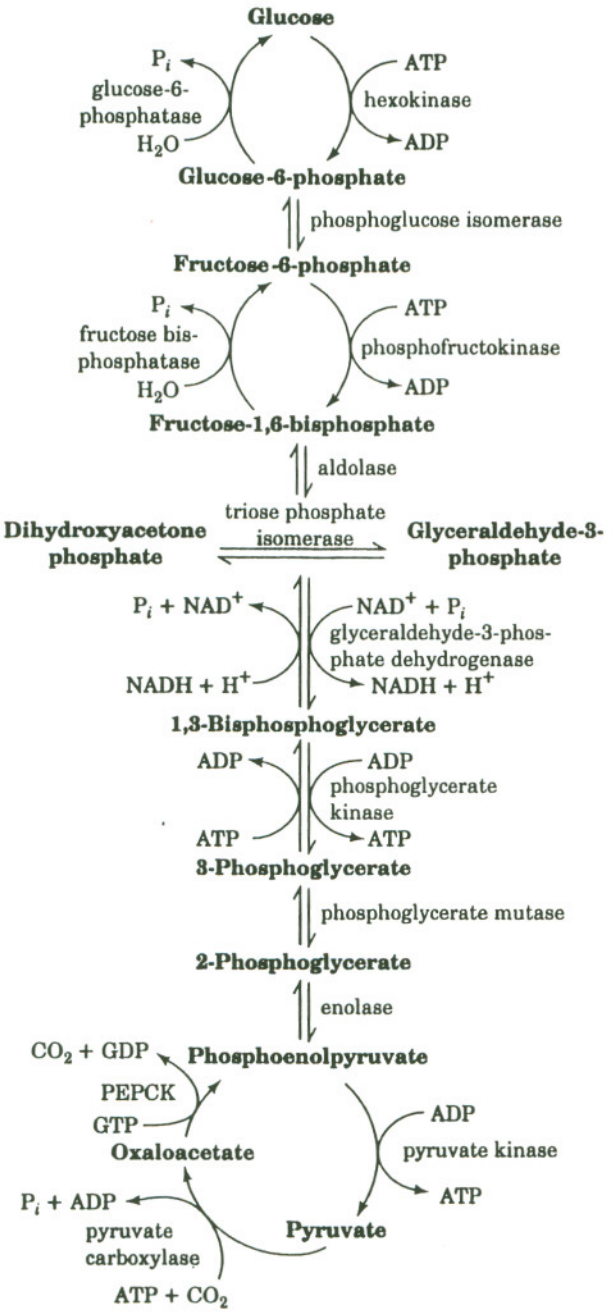


Figure 15-22. Key to Metabolism. **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium.

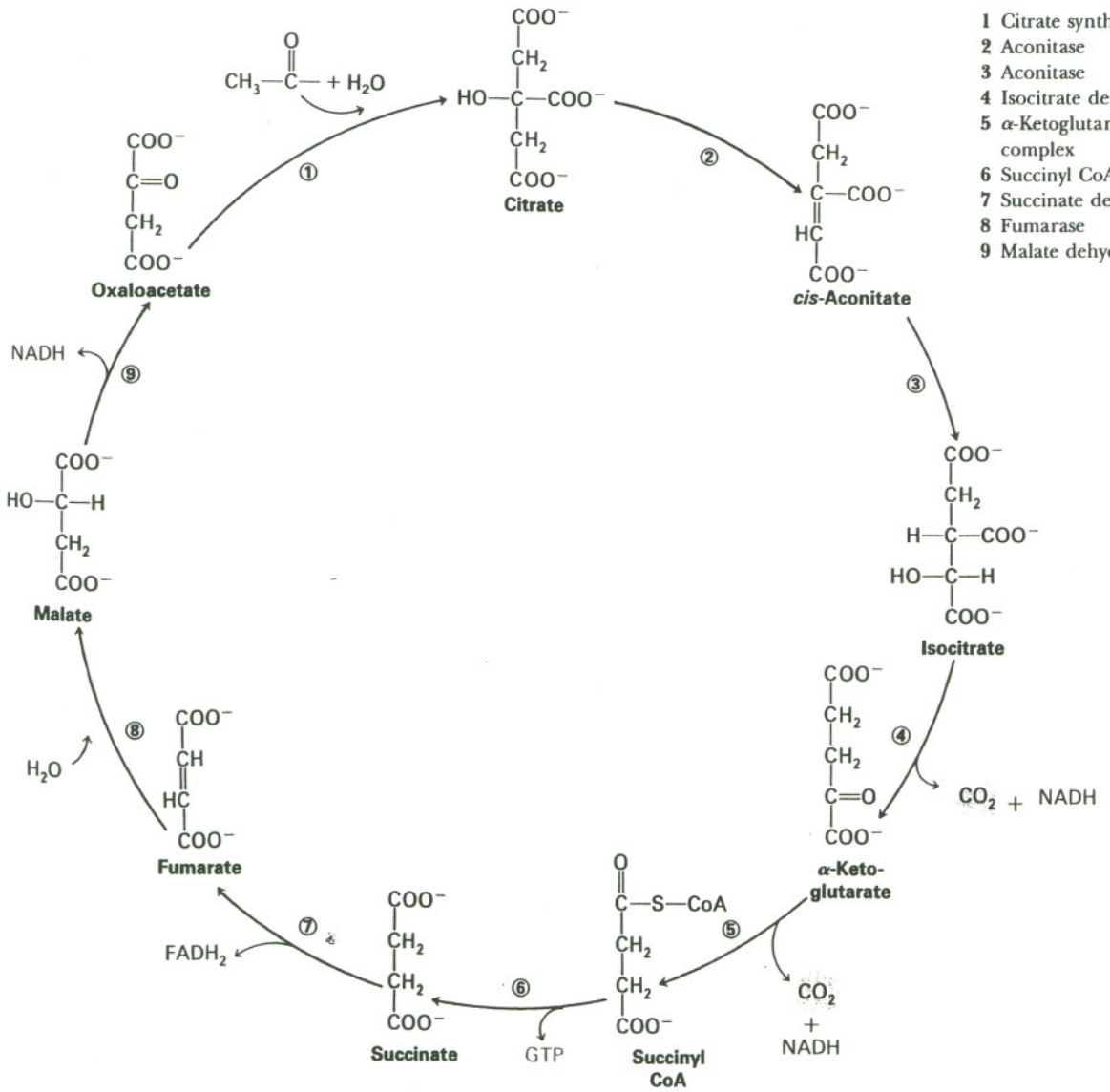


Figure 20-5
Citric acid cycle. This series of reactions is catalyzed by the following enzymes, as numbered in the diagram:

1 Citrate synthase
2 Aconitase
3 Aconitase
4 Isocitrate dehydrogenase
5 α-Ketoglutarate dehydrogenase complex
6 Succinyl CoA synthetase
7 Succinate dehydrogenase
8 Fumarase
9 Malate dehydrogenase

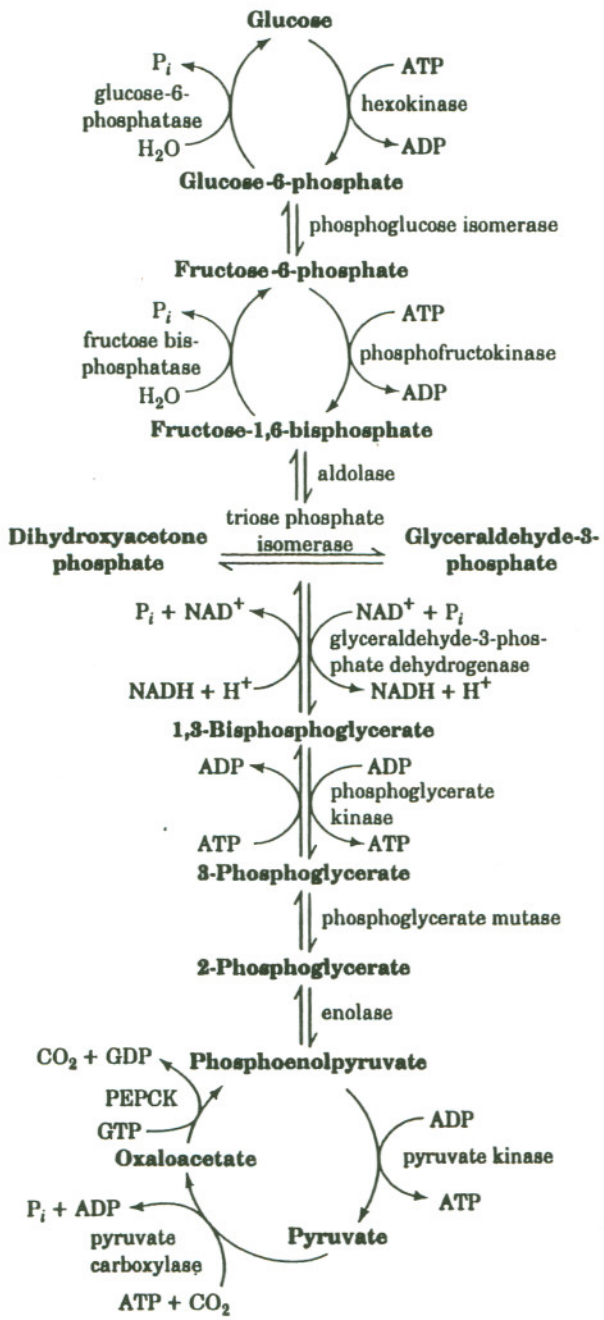


Figure 15-22. Key to Metabolism. **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium.